AI Powered E-Commerce Customer Intelligence System

Poori Report (Roman Urdu) – Har step ka "kya, kaise aur kyun"

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0. Project ek nazar mein

- **Maqsad:** Company ke raw database se shuru karke, data saaf karna, business analyse karna, churn aur sentiment ke models banana, aur sab kuch ek Streamlit app mein dikhana.
- **Pipeline (tarteeb):**
  1. Raw DB → 2. Cleaning → 3. Clean DB → 4. SQL queries → 5. EDA charts → 6. Churn ML → 7. Neural Network → 8. NLP Sentiment → 9. Saved models → 10. Streamlit app → 11. GitHub + Cloud deploy
- **Data:** 4 tables, total 100,000 rows

Table	Rows	Kya hai
customers	8,000	Customer profile, city, membership
products	1,000	Product, category, brand, price
orders	65,000	Transactions, discount, delivery, return
reviews	26,000	Rating aur review text

- **Project ki files:**

File	Kaam
`hackathon.ipynb`	Cleaning, EDA, ML, DL, NLP ka poora kaam
`queries.sql`	5 required SQL queries
`app.py`	Streamlit app (3 pages)
`ecommerce_clean.db`	Saaf kiya hua database (app aur SQL isi se chalte hain)
`models/churn_model.pkl`	Saved churn model
`models/sentiment_model.pkl`	Saved sentiment model
`requirements.txt`	Cloud ke liye libraries ki list
`README.md`	Summary + 3 business insights

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1. Notebook ka pehla cell: Libraries aur Database load

1.1 Libraries kyun import ki

Library	Kyun chahiye
`sqlite3`	Database se connect karne ke liye (rule: SQLite3 use karna hai)
`os`	Files check/delete karne ke liye
`joblib`	Trained model ko file mein save/load karne ke liye
`pandas`	Tables (DataFrames) par kaam
`numpy`	Numbers ka kaam
`matplotlib`	Charts banane ke liye
`train_test_split`	Data ko train aur test mein todne ke liye
`Pipeline / make_pipeline`	Kayi steps (scaling + model) ko ek saath jorna
`StandardScaler`	Numbers ko ek scale par lana
`LogisticRegression, RandomForestClassifier, MLPClassifier`	Churn ke 3 models
`TfidfVectorizer`	Text ko numbers mein badalna
`accuracy/precision/recall/f1/roc_auc/confusion_matrix/classification_report`	Models ko check karne ke metrics

1.2 Database load

- `RAW = "ecommerce_hackathon.db"` → original database (isko kabhi change nahi karte)
- `CLEAN = "ecommerce_clean.db"` → saaf copy jo hum khud banayenge
- `sqlite3.connect(RAW)` → database se connection
- `pd.read_sql("SELECT * FROM table", con)` → har table DataFrame mein
- **CSV kyun nahi banayi?** Rule 2: poora database manually CSV mein convert nahi karna.
- **Shape aur columns print kyun?** Task A ka pehla point hai (har table ki shape aur columns dikhana).

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2. Task A: Data Quality Summary (cleaning se PEHLE)

2.1 Summary cell kaise kaam karta hai

- Yeh cell **raw database se dobara data parhta hai** (variables `c, p, o, r`).
- **Kyun?** Agar memory wale cleaned tables par chalayen to sab 0 aayega. Raw se parhne par hamesha asli masle dikhte hain, chahe cell kisi bhi order mein chalayein.
- Har check ek line mein hai (count nikalta hai).

2.2 Kya masle mile (raw data mein)

Table	Masla	Count	Nuqsan agar theek na karein
customers	Age missing	100	Model/average galat ho jayega
customers	City ke spelling alag (`KARACHI, ' Lahore '`)	70	Ek shehar chart mein kayi hisson mein bat jata hai
customers	Duplicate email	35	Shayad data entry ghalti ya alag customers
products	Brand missing	20	Row ki info adhoori
products	Category ke spelling alag	18	Category chart mein double bars
orders	Invalid order_date (`unknown, 31-02-2025, 2026/13/01, 2025-00-14`)	30	Monthly trend aur churn mein date chahiye
orders	Quantity = 0	45	Zero quantity ka matlab sale hui hi nahi
orders	Unit price ≤ 0	35	Revenue galat (negative)
orders	Payment method missing	55	Analysis mein NaN
orders	Delivery days missing	40	Churn feature mein NaN
orders	Delivery days > 7	434	Sust delivery (shayad outlier)
orders	Discount > 30%	15	Bohat zyada discount

reviews	Rating 1-5 se bahar (0 aur 6)	25	Mumkin hi nahi rating
reviews	Review text khali/missing	90	NLP ke liye bekaar
reviews	Invalid review_date	16	Date parse nahi hoti
sab tables	Poori row duplicate	0	Koi masla nahi

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3. Cleaning: har step ka "kya aur kyun"

3.1 Step by step

Step	Kya kiya	Kyun
City	`.str.strip().str.title()`	Space hataye aur `karachi` / `KARACHI` ko `Karachi` banaya → ek shehar = ek naam
Age missing	**Median** se bhara	Median outliers se mutasir nahi hota; rows delete karne se customers kho jate
Category	`.str.strip().str.title()`	City wali wajah
Brand missing	`"Unknown"` likha	Row rakhi, nayi jhooti info nahi banayi
Order date	`pd.to_datetime(..., errors="coerce")`	Ghalat dates khud `NaT` (khali) ban jati hain
Invalid date wali rows	**Delete** (30)	Bina date ke order kisi mahine/window mein rakha nahi ja sakta
Quantity = 0	**Delete** (45)	Zero quantity ka matlab koi sale nahi
Negative/zero price	**Catalog price se replace** (35)	Delete nahi kiya kyun ke order ki price catalog ke bohat qareeb hoti hai; asli price pata hai
Payment method missing	`"Unknown"`	Row rakhi, sirf label diya
Delivery days missing	**Median** se bhara	Churn feature mein NaN nahi chalta; median safe hai
Rating 0 aur 6	**Delete** (25)	Rating sirf 1-5 hoti hai; asli value pata nahi
Review text missing	`""` (khali string)	Baad mein NLP step khud khali text hata dega
Review date	`errors="coerce"`	Date kahin use nahi hoti, to row delete nahi ki

3.2 Jo JAAN BUJH KAR NAHI hataya

- **Duplicate email (35):** alag alag customers hain; delete karte to unke orders bhi chale jate. Sirf flag kiya.
- **Delivery > 7 din (434):** sust lekin mumkin; yeh asli information hai.
- **Discount > 30% (14 bache):** kam hai lekin mumkin hai (sale/offer).
- **Rule 4:** "bina wajah ke delete nahi karna." Is liye har delete/fill ka reason upar likha hai.

3.3 Cleaning ke baad shapes

Table	Pehle	Baad
customers	8,000	8,000
products	1,000	1,000
orders	65,000	64,925 (30 invalid date + 45 zero quantity = 75 delete)
reviews	26,000	25,975 (25 ghalat rating delete)

3.4 Clean database kyun banayi

- Cleaned tables ko **naye file** `ecommerce_clean.db` mein save kiya (`to_sql(..., if_exists="replace")`).
- **Wajah 1:** SQL queries saaf data par chalni chahiyen.
- **Wajah 2:** Streamlit app isi file se parhegi.
- **Wajah 3:** Original raw database safe rehta hai.
- **Dates ko text (`YYYY-MM-DD`) mein save kiya** taake SQL ka `strftime` sahi chale.

3.5 Cleaning ke baad dobara check

- Jo masle theek ho sakte the woh **0** ho gaye.
- Bache hue (duplicate email 35, delivery>7 434, discount>30% 14, khali review text 90, invalid review date 16) **jaan bujh kar** rakhe hain (upar wajah di hai).
- **Judges ko dikhane ka tareeqa:** "Pehle" aur "baad" dono summary dikhao.

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4. Task B: SQL Business Analysis (`queries.sql`)

4.1 Net revenue ka formula

```
net revenue = quantity × unit_price × (1 − discount)
```

- **Kyun?** Customer ne discount ke baad paise diye, is liye asli kamai discount ke baad wali hai.

4.2 SQL ke keywords jo use hue

Keyword	Matlab
`SELECT`	Kaunse columns chahiye
`SUM()`, `COUNT()`	Jama karna, ginti karna

`JOIN ... ON`	Do tables ko common ID se jorna (orders + customers)
`GROUP BY`	Group bana kar har group ka hisab
`ORDER BY ... DESC`	Bare se chote ki taraf sort
`LIMIT`	Sirf top N rows
`ROUND()`	Decimal kam karna
`strftime('%Y-%m', date)`	Date se "saal-mahina" nikalna

4.3 Paanch queries

#	Query	Kya karti hai	Natija
Q1	Total net revenue	Poore orders ka net revenue `SUM`	**PKR 936.47 million**
Q2	Top 10 customers	orders + customers join, customer ke hisab se group, spending se sort, `LIMIT 10`	#1 Shahzaib Abbasi (Lahore), 43 orders, ~1.70M
Q3	Category-wise	orders + products join; revenue, order count, quantity	Electronics 656.6M (sab se zyada)
Q4	Monthly trend	`strftime` se mahina nikal kar group	Revenue lagatar barh raha hai; peak July 2026 (83.6M)
Q5	Top 5 products	orders + products join, product ke hisab se group	Samsung Headphones Classic 117.3M, Anker Headphones Classic 71.2M, ...

- **SQL kyun?** Rule 3: business queries SQL mein likhni hain; pandas sirf nikalne ke baad.
- **Notebook mein kaise chalti hain:** `queries.sql` file ko parh kar `;` par torte hain aur har query `read_sql` se chalate hain.

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5. Task C: EDA (4 charts)

5.1 Data kaise tayyar kiya

- Ek SQL query se `orders + customers + products` ko join karke `df` banaya (city, category, age, membership ek jagah).
- `revenue` column SQL mein hi banaya (`quantity × unit_price × (1 − discount)`).
- `parse_dates=["order_date"]` taake date asli date ban jaye.

5.2 Chaar charts

Chart	Kya dikhata hai
Monthly Revenue (line)	Waqt ke saath kamai

Monthly Revenue (line)	Waqt ke saath kamai
Revenue by Category (bar)	Kaunsi category zyada kamati hai
Revenue by City (bar)	Kaunsa shehar zyada kamata hai
Return Rate by Category (bar)	Kis category mein zyada maal wapas aata hai

5.3 Business insights (sirf dekhna nahi, "kyun zaroori")

1. **Electronics ~70% revenue** (656M / 936M), sirf 11.5k orders se. Do headphones products akele ~20% revenue dete hain.
   - *Matlab:* concentration risk. Stock-out ya supplier masla ho to kamai gir sakti hai. Stock aur backup supplier zaroori.
2. **Revenue barh raha hai magar August 2026 mein ~12% gira** (83.6M → 73.2M).
   - *Matlab:* ek mahina trend nahi, lekin check karna chahiye (season? stock? discount?).
3. **Fashion mein return rate sab se zyada (~11%)** aur orders sab se zyada (12.2k) magar revenue sirf ~6.6%.
   - *Matlab:* har return se shipping aur refund ka kharcha, margin kam. Size guide aur behtar photos se returns kam ho sakte hain. Groceries/Books sirf ~2%.
4. **Karachi (26%) + Lahore (18%) = 44% revenue.**
   - *Matlab:* in shehron mein delivery aur marketing sab se zyada asar karte hain; Islamabad jaise shehar growth ka mauqa.
5. **Churn recency aur frequency se chalta hai** (Random Forest mein `days_since_last_order` aur `total_orders` ~28-29% importance har ek).
   - *Matlab:* "X din se khareeda nahi" alert sab se saada retention trigger.

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6. Task D: Customer Churn (Machine Learning)

6.1 Setup (brief ke mutabiq)

- **Features:** sirf **31 May 2026 tak** ke orders se.
- **Target window: 1 June – 31 August 2026.**
- **Kaun se customers:** wo jinhon ne 31 May tak kam az kam ek order kiya (6,536 customers).
- **Churn = 1** agar target window mein koi purchase nahi; warna **0.**

6.2 Features (8)

Feature	Kaise bana	Kyun
total_orders	order ginti	Kitna aksar kharidta hai
total_spending	revenue ka sum	Kitna paisa kharch kiya
avg_order_value	spending ÷ orders	Aik order ka average size
days_since_last_order	31 May − aakhri order ki date	Kitni der se nahi khareeda (recency)

return_rate	`returned` ka mean	Kitna maal wapas kiya
avg_delivery_days	delivery days ka mean	Delivery ka tajurba
age	customers table se	Customer profile
membership_type	customers table se, number mein (Standard=0, Silver=1, Gold=2, Premium=3)	Tiers natural order mein hain, to number theek hai (OneHotEncoder ki zarurat nahi)

6.3 Data leakage se bachao (Rule 5)

- Features mein **sirf 31 May tak ka data**; target window ka koi order feature mein nahi.
- Target (`churn`) alag se window ke orders se banaya.
- **Scaler pipeline ke andar** hai, to sirf train data se seekhta hai (test ka koi andaza nahi).

6.4 Train/Test split

- `test_size=0.2` → 80% train, 20% test (test = 1,308 customers).
- `stratify=y` → train aur test dono mein churn ka tanasub barabar (49%).
- `random_state=42` → har baar wohi nateeja.

6.5 Models aur unki settings

Model	Setting	Kyun
Logistic Regression	`StandardScaler` + `max_iter=1000`	Saada, samajhne mein asaan; scaling zaroori kyun ke total_spending ke numbers lakhon mein hain aur age sirf 18-65
Random Forest	`n_estimators=200`, `max_depth=8`	200 darakht; depth limit se overfitting kam

- Churn rate 49% hai (classes barabar), is liye `class_weight` ki zarurat nahi.

6.6 Metrics asaan alfaaz mein

Metric	Matlab	Churn mein
Accuracy	Kitne sahi andaze (total)	Akeli kaafi nahi
Precision	Jinko model ne "churn" kaha, un mein se kitne sach mein churn hue	Ghalat alarm kitne
Recall	Jo sach mein churn hue, un mein se kitne pakre	**Sab se zaroori:** churner chhoot jana mehnga hai
F1	Precision aur Recall ka balance	Ek number mein khulasa
Confusion	TN, FP, FN, TP ki ginti	Kahan ghalti hui

Matrix		
ROC-AUC	Model churner ko stayer se upar rakhne mein kitna acha (0.5 = andaza, 1.0 = perfect)	Threshold se azad

6.7 Natije (test data)

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.724	0.675	0.844	0.750	0.785
Random Forest	0.722	0.674	0.839	0.748	0.789

- **Confusion matrix** `[[TN FP],[FN TP]]`:
  - Logistic Regression: `[[406, 261],[100, 541]]`
  - Random Forest: `[[407, 260],[103, 538]]`
- 51% baseline (hamesha "stay" bolna) ke muqable mein 72% accuracy asli behtari hai.

6.8 Kaunsa model aur kyun

- Dono ka nateeja lagbhag **ek jaisa**.
- **Maine Logistic Regression chuna kyun ke:**
  - Same performance, lekin saada aur tez
  - Samjhana asaan (har feature ka coefficient)
  - Model file chhoti, deployment halka
- **Recall par zor kyun:** churner ko miss karna (customer kho dena) ghalat alarm (ek email/discount) se zyada mehnga hai. LR 84% churners pakarta hai.

6.9 Model save kyun

- `joblib.dump(lr, "models/churn_model.pkl")` → app bina dobara train kiye prediction deti hai.
- Rule 6: hard-coded predictions nahi, saved model use karna hai.

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7. Task E: Deep Learning (Neural Network)

- **Architecture:** `32 → 16 → 1`
  - 2 hidden layers (32 aur 16 neurons), `relu` activation
  - Output: 1 neuron (sigmoid), loss: log-loss (binary classification)
  - `early_stopping=True` → overfitting se pehle ruk jata hai
  - Scaling ke saath (`make_pipeline(StandardScaler(), MLPClassifier(...))`)
- **Same split, same metrics** (fair muqabla).

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.72	0.67	0.84	0.75	0.79

Random Forest	0.72	0.67	0.84	0.75	0.79
Neural Network	0.72	0.68	0.80	0.74	0.79

- **Kya DL worth it tha? Nahi.**
  - Performance lagbhag same, recall thori kam (0.80 vs 0.84)
  - Sirf 8 saada features aur ~6,500 customers; itne data mein NN ko seekhne ke liye gehri complexity nahi mili
  - Samjhana aur tune karna mushkil → extra complexity ka faida nahi

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8. Task F: NLP Sentiment Analysis

8.1 Steps

Step	Kya	Kyun
Khali text hataya	`WHERE TRIM(review_text) != ''`	Khali text se kuch seekh nahi sakte
Ghalat rating	Pehle hi cleaning mein hata di	Labels sahi hon
Labels	Rating 1-2 = Negative, 3 = Neutral, 4-5 = Positive (`pd.cut`)	Brief ke mutabiq
Duplicate sentences hataye	`drop_duplicates("review_text")`	Wohi jumla train aur test dono mein na ho, warna test jhoota-asan ho jata
TF-IDF	`ngram_range=(1,2)`, `min_df=2`	Alfaaz aur do-lafzi jore ("not recommended") ko numbers banata hai; `min_df=2` bohat rare alfaaz hata deta hai
Stop words **nahi** hataye	**-**	"not" jaise alfaaz ka matlab badal dete hain ("not recommended")
Model	Logistic Regression (`class_weight="balanced"`)	Brief ke mutabiq; Neutral/Negative kam hain to balance
Lowercase/punctuation	TF-IDF khud karta hai	Alag code ki zarurat nahi

8.2 Data ka hisaab

- 25,885 reviews (khali text ke baad), lekin **sirf 4,484 unique jumle**
- Duplicate hatane ke baad: Positive 2,160, Neutral 1,387, Negative 937
- Test set: 897 reviews

8.3 Natija

- Accuracy, Precision, Recall, F1 sab **1.00.**

8.4 Limitation (zaroor likhein)

- **Reviews template se bani hain:** sirf ~4,500 alag jumle ~26,000 reviews mein, aur har sentiment ke apne alag alfaaz hain.
- Is liye 1.00 score **asli duniya mein nahi chalega**. Mera test: "terrible and broke in two days" ko model ne **Positive** kaha (naye alfaaz ka tajurba nahi).
- Dusri limitation: labels star rating se bane hain, jo hamesha text se match nahi karti.
- Roman Urdu reviews (jaise "bilkul theek nikla") ko model sirf aam alfaaz ki tarah leta hai.
- **Judges ko 1.00 ko kamyabi nahi, data ki khasiyat bata kar dikhayen.**

8.5 Save

- `joblib.dump(sent, "models/sentiment_model.pkl")` : pipeline mein TF-IDF aur Logistic Regression dono hain, to app seedha asli text le sakti hai.

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9. Task G: Streamlit App (app.py)

9.1 Shuruaat

- `BASE = Path(__file__).parent` → **relative paths**; Cloud par bhi chalega (Rule: relative paths).
- `@st.cache_data` → SQL ka natija yaad rakhta hai (app tez).
- `@st.cache_resource` → model sirf ek baar load hota hai.
- Sidebar radio se 3 pages.

9.2 Page 1: Dashboard

- **4 KPI cards:** Total revenue, total orders, total customers, return rate.
- **Charts:** monthly revenue (line), category revenue (bar), city revenue (bar).
- **Tables:** top 10 customers, top 5 products.
- **Sab kuch SQL se** (brief ko kam az kam 3 SQL outputs chahiye; yahan 7+ queries hain).

9.3 Page 2: Churn Prediction

- **Inputs:** total orders, total spending, days since last order, return rate, average delivery days, age, membership.
- `avg_order_value` khud banta hai (spending ÷ orders), taake user ko do baar na dena pare.
- Membership ko wohi number coding (0-3) jo notebook mein thi.
- Columns ko `model.feature_names_in_` ke order mein lagaya (training jaisa).
- **Output:** "Likely to CHURN" ya "Likely to STAY" aur **probability** (`predict_proba`) + progress bar.
- Tested nateeja: 10 orders → ~31% churn; 2 orders → ~58% churn (kam orders = zyada churn khatra, jo data ke mutabiq hai).

9.4 Page 3: Sentiment Analysis

- Text box mein review likhein → saved model → **Negative/Neutral/Positive** + har class ki probability.
- Khali text par warning.
- Neeche note: model template-style reviews par train hua hai, to ajeeb alfaaz ghalat ho sakte hain.

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10. Task H: GitHub aur Streamlit Cloud

10.1 GitHub par kya upload karna hai

- `hackathon.ipynb`, `queries.sql`, `app.py`, `README.md`, `requirements.txt`
- `ecommerce_clean.db` (**zaroori**, app isi se chalti hai) aur `ecommerce_hackathon.db`
- `models/` folder (dono `.pkl` files)
- Dono databases chhote hain (11 MB aur 8 MB), GitHub par theek.

10.2 `requirements.txt` mein version kyun

- Notebook ka aakhri cell `scikit-learn==...` aur `joblib==...` khud likhta hai (aap ke apne version).
- **Kyun?** Pickle model usi sklearn version par load hota hai jis par train hua; Cloud par version alag ho to masla.

10.3 Deploy ke steps

1. GitHub par naya repo banao, sari files upload karo.
2. `share.streamlit.io` par login (GitHub se).
3. "New app" → apna repo chuno → Main file: `app.py`.
4. Deploy dabao, 2-3 minute intezaar.
5. Live link kholkar **teeno pages test** karo.
6. Link `README.md` mein likho aur submit karo.

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11. Important numbers (ek jagah)

Cheez	Value
Total net revenue	PKR 936.47 million
Valid orders	64,925
Total customers	8,000
Overall return rate	~6.7%
Electronics ka hissa	~70%
Karachi + Lahore ka hissa	~44%
Churn rate	49% (6,536 customers)

Best churn model	Logistic Regression: Recall 0.84, F1 0.75, AUC 0.79
Neural Network	Recall 0.80, F1 0.74, AUC 0.79
Sentiment score	1.00 (template data ki wajah se)

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12. Judges ke mumkina sawal (taiyari)

1. **Data leakage kaise roki?** Features sirf 31 May tak; target baad ki window se; scaler pipeline ke andar sirf train se seekhta hai.
2. **Orders kyun delete kiye?** Sirf 75: 30 ki date ghalat (waqt ka pata nahi), 45 mein quantity 0 (sale nahi hui). Baqi masle fill/repair kiye.
3. **Negative price kyun delete nahi ki?** Catalog price maujood thi, to repair kar di.
4. **Median kyun, mean kyun nahi?** Outliers mean ko khenchte hain, median nahi.
5. **Accuracy kaafi kyun nahi?** Ghalat ginti ke qisam (FP/FN) nahi batati; churn mein FN mehnga hai, is liye recall/F1 dekhe.
6. **Logistic Regression kyun chuna?** Nateeja Random Forest jaisa, lekin saada, samajhne layak, halka.
7. **Neural Network kyun na use kiya?** Behtari nahi mili, complexity zyada.
8. **Sentiment 1.00 kyun?** Reviews template-based hain (4,484 unique jumle), asli duniya mein score kam hoga.
9. **Net revenue ka formula?** quantity × unit_price × (1 − discount).
10. **App hard-coded to nahi?** Nahi, saved `.pkl` models se predict karti hai aur dashboard SQL se chalta hai.

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13. Is project ki kamzoriyan (imandari se)

- Churn ki definition "3 mahine mein khareedari nahi" hai; kuch customers sirf kam kharidte hain, asli "chhor gaye" nahi.
- `membership_type` customer ki mojooda halat hai, 31 May ki nahi; thora sa mismatch ho sakta hai.
- Sentiment data synthetic/template hai.
- Aakhri mahina (Aug 2026) ki girawat ek mahine ka data hai, trend nahi.